A SURJECTION BETWEEN FINITE LOCALLY FREE MODULES OF THE SAME RANK IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local triviality of a finite locally free module of a fixed rank. The author's section on locally free sheaves, in which the counted statement occurs. Reduces the statement to a map between finite free modules over the ring of sections of an open subset.
- The algebraic statement that a surjective endomorphism of a finite free module of fixed rank is an isomorphism. Invoked inside the counted proof at source lines 2386-2388 as a cross-chapter pointer. Upgrades the surjectivity of the map on sections to injectivity, which is what remains to be shown.

Source reference. Original reference algebra-lemma-fun: sources/stacks-modules/pinned-original-source.tex, source lines 2387–2387. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-fun, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $r \geq 0$. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a map of finite locally free $\mathcal{O}_X$-modules of rank r. Then φ is an isomorphism if and only if φ is surjective.*

Proof. Assume φ is surjective. Pick $x \in X$. There exists an open neighbourhood U of x such that both $\mathcal{F}|_U$ and $\mathcal{G}|_U$ are isomorphic to $\mathcal{O}_U^{\oplus r}$. Pick lifts of the free generators of $\mathcal{G}|_U$ to obtain a map $\psi : \mathcal{G}|_U \to \mathcal{F}|_U$ such that $\varphi|_U \circ \psi = \mathrm{id}$. Hence we conclude that the map $\Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{G})$ induced by φ is surjective. Since both $\Gamma(U, \mathcal{F})$ and $\Gamma(U, \mathcal{G})$ are isomorphic to $\Gamma(U, \mathcal{O}_U)^{\oplus r}$ as an $\Gamma(U, \mathcal{O}_U)$-module we

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may apply Algebra, Lemma algebra-lemma-fun to see that $\Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{G})$ is injective. This finishes the proof. $\square$